

**Cayley — Collected Mathematical Papers, Vol. VII**

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Next, the case  $n = 2$ ,  $m_1 = 2$ ,  $h_1 = 0$ ,  $a_1 = 1$ : the reciprocal transformation is of the order  $n' = 2$ ; it is evidently not of the form above considered (for this would make the original transformation to be of the order 3). Hence, assuming (as it seems allowable to do) that the principal system does not contain any multiple point or curve, the reciprocal transformation will be of the same form as the original one, viz., we shall have  $n' = 2$ ,  $m'_1 = 2$ ,  $h'_1 = 0$ ,  $a'_1 = 1$ .

87. Considering next the cubic transformations, or those belonging to  $n = 3$ ; in the case  $m_1 = 4$ ,  $h_1 = 4$ ,  $a_1 = 2$ , the reciprocal transformation is of the order  $9 - 4 = 5$ ; and in the case  $m_1 = 5$ ,  $h_1 = 5$ ,  $a_1 = 1$ , the reciprocal transformation is of the order  $9 - 5 = 4$ : I do not consider these cases. But  $m_1 = 6$ ,  $h_1 = 7$ ,  $a_1 = 0$ , the reciprocal transformation is of the order  $-9 + 6 = 3$ ; and assuming (as seems allowable) that the principal system does not contain any multiple point or curve, it must be of the same form as the original transformation, that is, we must have  $n' = 3$ ,  $m'_1 = 6$ ,  $h'_1 = 7$ ,  $a'_1 = 0$ .

88. The transformations to be studied are thus, 1° The quadri-quadric transformation  $n = 2$ ,  $m_1 = 2$ ,  $h_1 = 0$ ,  $a_1 = 1$ , and  $n' = 2$ ,  $m'_1 = 2$ ,  $h'_1 = 0$ ,  $a'_1 = 1$ ; the principal system in each space consists of a point and of a conic (which may be a pair of intersecting lines); and the surfaces are quadrics. 2° The quadri-cubic transformation  $n = 2$ ,  $m_1 = 1$ ,  $h_1 = 0$ ,  $a_1 = 3$ , and  $n' = 3$ ,  $a'_1 = 0$ ,  $h'_2 = 3$ ,  $m'_2 = 1$ ,  $h'_2 = 0$ ; in the first space the principal system consists of three points and a line, and the surfaces are quadrics: in the second figure, the principal system consists of three simple lines and a double line; and the surfaces are cubic surfaces passing through this principal system, that is, they are cubic scrolls. 3° The cubo-cubic transformation  $n = 3$ ,  $a_1 = 0$ ,  $m_1 = 6$ ,  $h_1 = 7$ , and  $n' = 3$ ,  $a'_1 = 0$ ,  $m'_1 = 6$ ,  $h'_1 = 7$ ; in each space the principal system is a sextic curve with seven apparent double points (but there are different cases to be considered according as the sextic curve does or does not break up into inferior curves), and the surfaces are cubic surfaces through the sextic curve.

*The Quadri-quadric Transformation between Two Spaces.*

89. Starting from the equations  $x' : y' : z' : w' :: X : Y : Z : W$ , we have here  $X = 0$ , &c., quadric surfaces passing through a given point and a given conic (which may be a pair of intersecting lines). Take  $x = 0$ ,  $y = 0$ ,  $z = 0$  for the coordinates of the given point;  $w = 0$  for the equation of the plane of the conic; the conic is then given as the intersection of this plane by a cone having the given point for its vertex; or say the equations of the conic are  $w = 0$ ,  $(a, \dots | x, y, z)^2 = 0$ ; the general equation of a quadric through the point and conic is  $w(\alpha x + \beta y + \gamma z) + h(a, \dots | x, y, z)^2 = 0$ ; and it hence appears that the equations of the transformation may be taken to be

$$x' : y' : z' : w' = xw : yw : zw : (a, \dots | x, y, z)^2;$$

these give at once a reciprocal system of the same form; viz., the two sets are

$$x' : y' : z' : w' = xw : yw : zw : (a, \dots | x, y, z)^2,$$

and

$$x : y : z : w = x'w' : y'w' : z'w' : (a, \dots | x', y', z')^2.$$

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90. The Jacobian of the first space is at once found to be

$$w^2(a, \dots | x, y, z)^2 = 0;$$

that of the second space is of course

$$w'^2(a, \dots | x', y', z')^2 = 0.$$

The two spaces are similar to each other; we may say that there is in each of them a principal point and a principal conic; that the plane of the conic is the principal plane, and the cone having its vertex at the point and passing through the conic is the principal cone. To the principal point of either space corresponds any point whatever in the principal plane of the other space; and conversely. More definitely, the points of the one principal plane and the infinitesimal elements of direction through the principal point of the other space correspond according to the equations  $x : y : z = x' : y' : z'$ . To any point on the principal conic of either space corresponds in the other space, not a mere element of direction through the principal point of the other space, but a line of the principal cone; that is, to the points of the principal conic of the one space correspond the lines of the principal cone of the other space. The Jacobian of either space, consisting of the principal plane twice, and of the principal cone, is thus the principal counter-system of the other space.

91. {Writing  $(a, \dots | x, y, z)^2 = x^2 + y^2 + z^2$ ,  $w = w' = 1$ , the equations of transformation become

$$x' : y' : z' : 1 = x : y : z : x^2 + y^2 + z^2,$$

and

$$x : y : z : 1 = x' : y' : z' : x'^2 + y'^2 + z'^2,$$

or, what is the same thing, if for shortness

$$x^2 + y^2 + z^2 = r^2, \quad x'^2 + y'^2 + z'^2 = r'^2,$$

the equations are

$$x' = \frac{x}{r^2}, \quad y' = \frac{y}{r^2}, \quad z' = \frac{z}{r^2}; \quad \text{and} \quad x = \frac{x'}{r'^2}, \quad y = \frac{y'}{r'^2}, \quad z = \frac{z'}{r'^2},$$

whence also  $rr' = 1$ ; this is the well known transformation by reciprocal radius vectors.}

92. The principal conic may be a pair of intersecting lines; taking its equations to be  $w = 0$ ,  $xy = 0$ , the equations of transformation here become

$$x' : y' : z' : w' = yw : xw : xy : zw,$$

and

$$x : y : z : w = y'w' : x'w' : x'y' : z'w'.$$

There is no difficulty in the further development of the theory.

### *The Quadri-cubic Transformation between Two Spaces.*

93. It will be convenient to have the unaccented letters  $(x, y, z, w)$  referring to the cubic surfaces. I will therefore take the quadric surfaces in the second figure; viz., I will start from the equations  $x : y : z : w = X' : Y' : Z' : W'$ , where  $X' = 0$ ,  $Y' = 0$ ,  $Z' = 0$ ,  $W' = 0$  are quadric surfaces passing through three fixed points (say the principal points) and through a fixed line (say the principal line) in the second figure. Taking  $x' = 0$ ,  $y' = 0$  for the planes passing through the principal line and through two of the principal points respectively;  $z' = 0$  for the plane passing through the three principal points,  $w' = 0$  for an arbitrary plane passing through the first mentioned two principal points, the implicit factors of  $x'$ ,  $y'$ ,  $w'$  may be so determined that for the third principal point  $x' = y' = -w'$ . That is, we shall have

$$\begin{aligned} \text{for principal line} & \quad x' = 0, \quad y' = 0, \\ \text{for principal points} & \quad (x' = 0, \quad z' = 0, \quad w' = 0), \\ & \quad (y' = 0, \quad z' = 0, \quad w' = 0), \\ & \quad (x' = y' = -w', \quad z' = 0), \end{aligned}$$

and this being so, the equation of a quadric surface through the principal points and line will be

$$(\alpha x' + \beta y')z' + \gamma x'(y' + w') + \delta y'(x' + w'),$$

and the equations of transformation may be taken to be

$$x : y : z : w = x'z' : y'z' : x'(y' + w') : y'(x' + w').$$

94. Writing these in the extended form

$$x : y : z : w : z - w : x - y = x'z' : y'z' : x'(y' + w') : y'(x' + w') : z'(x' - y') : w'(x - y)$$

and forming also the equation

$$xy : (xw - yz) = z' : x' - y',$$

we at once derive the reciprocal system of equations

$$x' : y' : z' : w' = x(xw - yz) : y(xw - yz) : (x - y)xy : (z - w)xy,$$

so that this is a cubic transformation. And the cubic surface in the first space (corresponding to an arbitrary plane  $ax' + by' + cz' + dw' = 0$  of the second space) is  $(ax + by)(xw - yz) + c(x - y)xy + d(z - w)xy = 0$ ; viz., this is a cubic surface having the fixed double line  $(x = 0, y = 0)$ , the fixed simple lines  $(x = 0, z = 0)$ ,  $(y = 0, w = 0)$ , and  $(x - y = 0, z - w = 0)$ ; it has also the variable simple line  $(dz + cx = 0, dw + cy = 0)$ . The principal figure of the first space thus consists of the three simple lines  $(x = 0, z = 0)$ ,  $(y = 0, w = 0)$ ,  $(x - y = 0, z - w = 0)$ , and of the line  $(x = 0, y = 0)$ , a double line counting four times in the intersection of two of the cubic surfaces.

95. The cubic surface as having the double line  $(x = 0, y = 0)$  is a cubic scroll, and this line is the nodal directrix thereof; the line  $(dz + cx = 0, dw + cy = 0)$  is the simple directrix; the lines  $(x = 0, z = 0)$ ,  $(y = 0, w = 0)$ ,  $(x - y = 0, z - w = 0)$  are at once seen to be lines meeting each of these directrix lines; and they are generating lines of the scroll. To explain the generation of the scroll, observe that the section by any plane is a cubic curve having a given double point (viz., the intersection of the plane with the nodal directrix); and three other given points (viz., the intersections of

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the plane with the three generating lines respectively); this cubic also passes through the intersection of the plane with the simple directrix. Conversely, if the plane be assumed at pleasure, and if, taking for the simple directrix any line which meets the given generating lines, we draw a cubic as above, then the scroll is the scroll generated by a line which meets each of the directrix lines, and also the cubic.

If the plane be taken to pass through any generating line, then the cubic section breaks up into this line, and a conic; the conic does not meet the simple directrix, but it meets the nodal directrix; and any such conic will serve as a directrix; viz., the scroll is generated by the lines which meet the two directrix lines and the conic.

96. Any two scrolls as above meet in the three fixed generating lines, and in the nodal directrix counting four times; they consequently meet besides in a curve of the second order, which is a conic (one of the conics just referred to). In order to further explain the theory, suppose for a moment that the two scrolls had only a common nodal directrix; they would besides meet in a quintic curve; this curve would meet the nodal directrix in four points, viz., the points at which the two scrolls have a common tangent plane. Now if at any point of the nodal directrix the two scrolls have a common generating line, then the plane through this line and the nodal line is one of the two tangent planes of each scroll; that is, the scrolls have this plane for a common tangent plane. Hence, in the case of the common three generating lines, the points where these meet the nodal line are three of the four points just referred to; there remains therefore one point, which is the point where the conic meets the nodal line; through this point there are for each of the scrolls two generating lines; one of these for the first scroll, and one for the second scroll, lie in a plane with the nodal line; the other two determine the plane of the conic; and the tangent to the conic at its intersection with the nodal line is the intersection of the plane of the conic with the plane of the first-mentioned two generating lines.

97. Analytically we have the two equations

$$\begin{aligned} c(x - y)xy + (ax + by)(xw - yz) + d(z - w)xy &= 0, \\ c'(x - y)xy + (a'x + b'y)(xw - yz) + d'(z - w)xy &= 0; \end{aligned}$$

or, combining these equations so as to eliminate successively the terms in  $x(xw - yz)$  and  $y(xw - yz)$ , and for this purpose writing

$$(bc' - b'c, ca' - c'a, ab' - a'b, ad' - a'd, bd' - b'd, cd' - c'd) = (a, b, c, f, g, h),$$

and therefore

$$af + bg + ch = 0,$$

we have

$$\begin{aligned} h(x - y)x - c(xw - yz) - f(z - w)x &= 0, \\ -h(x - y)y + c(xw - yz) - g(z - w)y &= 0, \end{aligned}$$

and multiplying the first of these by  $c + g$  and the second by  $c - f$ , and adding, the whole divides by  $x - y$ , and the final result is

$$(c + g)(hx - fy) - (c - f)(ay + gx) = 0;$$

viz., this is the equation of the plane of the conic.

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98. Any two scrolls as above meeting in a conic, a third scroll will meet the conic in six points; but these include the point on the nodal directrix twice, and the points on the three fixed generating lines each once; there is left a single point of intersection, viz., this is the one variable point of intersection of the three scrolls which is in accordance with the theory.

99. For the Jacobian of the second space, we have

$$\begin{vmatrix} z' & . & x' & . \\ . & z' & y' & . \\ y' + w' & x' + w' & . & x' \\ . & . & x' + w' & 0 & y' \end{vmatrix} = 0;$$

that is,  $x'y'z'(x' - y') = 0$ ; viz.,  $z' = 0$  is the plane containing the three principal points; and  $x' = 0$ ,  $y' = 0$ ,  $x' - y' = 0$  are the planes which pass through the principal line and the three principal points respectively.

100. For the Jacobian of the first space, we have

$$\begin{vmatrix} 2xw - yz & -xz & -xy & x^2 \\ yw & xw - 2yz & -y^2 & xy \\ y(z - w) & x(z - w) & xy & -xy \\ (z - w)y & (z - w)x & xy & -xy \end{vmatrix} = 0;$$

that is,  $3x^2y(x - y)^2(xw - yz) = 0$ ; viz.,  $x = 0$ ,  $y = 0$ ,  $x - y = 0$  are the planes through the nodal directrix and the three fixed generators respectively (each plane therefore occurring twice); and  $xw - yz = 0$  is the quadric scroll generated by the lines which meet each of the three generators ( $x = 0$ ,  $z = 0$ ), ( $y = 0$ ,  $w = 0$ ), ( $x - y = 0$ ,  $z - w = 0$ ); this scroll passing also through the nodal directrix  $x = 0$ ,  $y = 0$ .

#### *The Cubo-cubic Transformation between Two Spaces.*

101. The principal system in the first space is a sextic curve with 7 apparent double points; but this curve may be either a single curve, or it may break up into inferior curves. I have not examined all the cases which may arise; but the two extreme cases are (A) The sextic curve breaks up into six lines, viz., two non-intersecting lines, and four other lines each meeting each of the two lines (this implies that no two of the four lines meet each other): here the two lines give 1 apparent double point, and the four lines give 6 apparent double points; total number is = 7, as it should be. (B) The curve is a proper sextic curve, with 7 apparent double points: this gives, as will be shown, the general lineo-linear transformation. The two cases are each of them symmetrical.

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(A) *The Principal System consists of Six Lines.*

102. Taking in the first space, for the equations of the two lines,  $(x = 0, y = 0)$  and  $(z = 0, w = 0)$ , and for the equations of the four lines,  $(x = 0, z = 0)$ ,  $(y = 0, w = 0)$ ,  $(x - y = 0, z - w = 0)$ ,  $(x - py = 0, z - qw = 0)$ , then, if the equations of transformation are taken to be

$$\begin{aligned} x' - py' : x' - y' : z' - qw' : z' - w' &= (x - py)(xw - yz) \\ &: (x - y)(qwx - pyz) \\ &: (z - qw)(xw - yz) \\ &: (z - w)(qwx - pyz); \end{aligned}$$

these lead conversely (see post, No. 104) to a like system,

$$\begin{aligned} x - py : x - y : z - qw : z - w &= (x' - py')M' \\ &: (x' - y')N' \\ &: M'(z' - qw') \\ &: (z' - w')N', \end{aligned}$$

where for shortness

$$\begin{aligned} M' &= p(q - 1)^2 x'w' - q(p - 1)^2 y'z' + (pq - 1)(p - q)y'w', \\ N' &= (q - 1)^2 x'w' - (p - 1)^2 y'z' + (pq - 1)(p - q)y'w'; \end{aligned}$$

or, as these are better written,

$$\begin{aligned} M' &= -q(p - 1)y'\{(p - 1)z' - (pq - 1)w'\} + p(q - 1)w'\{(q - 1)x' - (pq - 1)y'\}, \\ N' &= -(p - 1)y'\{(p - 1)z' - (pq - 1)w'\} + (q - 1)w'\{(q - 1)x' - (pq - 1)y'\}. \end{aligned}$$

Hence the principal system in the second plane is composed of the two non-intersecting lines  $(x' = 0, y' = 0)$ ,  $(z' = 0, w' = 0)$  and the four lines  $\{(p - 1)z' - (pq - 1)w' = 0, (q - 1)x' - (pq - 1)y' = 0\}$ ,  $(y' = 0, w' = 0)$ ,  $(x' - y' = 0, z' - w' = 0)$ ,  $(x' - py' = 0, z' - qw' = 0)$ , each meeting each of the two lines.

103. The Jacobian of the first space is

$$\begin{vmatrix} 2xw - yz - pyw, & -xz - pxw + 2pyz, & -y(x - py), & x(x - py) \\ 2qwx - pyz - qyw, & -pxz - qwx + 2pyz, & -py(x - y), & -x(x - y) \\ w(z - qw), & -z(z - qw), & xw - 2yz + qyw, & xz - 2qwx + qyz \\ qw(z - w), & -pz(z - w), & qwx - 2pyz + pyw, & qxz - 2qwx + pyz \end{vmatrix} = 0,$$

viz., this is  $xyzw(x - y)(x - py)(z - w)(z - qw) = 0$ , the equation of the planes each passing through one of the four lines and one of the two lines.

Similarly, the Jacobian of the second space is

$$\{(q - 1)x' - (pq - 1)y'\}\{(p - 1)z' - (pq - 1)w'\}(x' - y')(z' - w')(x' - py')(z' - qw')y'w' = 0;$$

viz., this is the equation of the eight planes each passing through one of the four lines and one of the two lines.

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104. To effect the foregoing transformation, writing

$$\begin{aligned} x' : y' : z' : w' &= (x - py)(xw - yz) \\ &: (x - y)(qwx - pyz) \\ &: (z - qw)(xw - yz) \\ &: (z - w)(qwx - pyz); \end{aligned}$$

or what will ultimately be the same thing, but it is more convenient for working with,

$$\begin{aligned}x' &= (x - py)(xw - yz), \\y' &= (x - y)(qxw - pyz), \\z' &= (z - qw)(xw - yz), \\w' &= (z - w)(qxw - pyz),\end{aligned}$$

these give

$$\begin{aligned}x - py &= M'x', \\x - y &= N'y', \\z - qw &= M'z', \\z - w &= N'w',\end{aligned}$$

where  $M'$ ,  $N'$  are quantities which have to be determined; and thence

$$\begin{aligned}(1 - p)x &= M'x' - pN'y', \\(1 - p)y &= M'x' - N'y', \\(1 - q)z &= M'z' - qN'w', \\(1 - q)w &= M'z' - N'w';\end{aligned}$$

whence also

$$\begin{aligned}(1 - p)(1 - q)(xw - yz) &= N'[\{(q - 1)x'w' - (p - 1)y'z'\}M' + (p - q)y'w'N'], \\(1 - p)(1 - q)(qxw - pyz) &= M'[-(p - q)x'z'M' + \{(pq - q)x'w' - (pq - p)y'z'\}N'];\end{aligned}$$

but we have

$$\frac{xw - yz}{qxw - pyz} = \frac{x'}{y'} \cdot \frac{x - py}{x - y} = \frac{x'}{y'} \cdot \frac{M'x'}{N'y'} = \frac{N'}{M'},$$

or, substituting,

$$\begin{aligned}M'\{(q - 1)x'w' - (p - 1)y'z'\} + N'(p - q)y'w' &= M'\{-(p - q)x'z'\} + N'\{(pq - q)x'w' - (pq - p)y'z'\}; \\M'\{(q - 1)x'w' - (p - 1)y'z' + (p - q)x'z'\} &= N'\{(pq - q)x'w' - (pq - p)y'z' - (p - q)y'w'\};\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}M' &= (pq - q)x'w' - (pq - p)y'z' - (p - q)y'w', \\N' &= (q - 1)x'w' - (p - 1)y'z' + (p - q)x'z';\end{aligned}$$

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viz.,  $M'$ ,  $N'$  having these values, the original equations

$$\begin{aligned}x' : y' : z' : w' &= (x - py)(xw - yz) \\&: (x - y)(pxw - qyz) \\&: (z - qw)(xw - yz) \\&: (z - w)(pxw - qyz),\end{aligned}$$

give

$$x - py : x - y : z - qw : z - w = M'x' : N'y' : M'z' : N'w'.$$

If, in these equations, in place of  $(x', y', z', w')$  we write  $(x' - py', x' - y', z' - qw', z' - w')$ , the new values of  $M'$ ,  $N'$  are found to be

$$\begin{aligned}M' &= p(q - 1)^2x'w' - q(p - 1)^2y'z' + (pq - 1)(p - q)y'w', \\N' &= (q - 1)^2x'w' - (p - 1)^2y'z' + (pq - 1)(p - q)y'w',\end{aligned}$$

and we have the formulae of No. 102.

(B) *The Principal System of a Proper Sextic Curve; the Lineo-linear Transformation between Two Spaces.*

105. I start with the lineo-linear transformation, and show that this is in fact a transformation such that the principal system in either space is a sextic curve with seven apparent double points. I do not attempt any formal proof, but assume that the lineo-linear transformation is the most general one which gives rise to such a principal system.

We have between  $(x, y, z, w)$ ,  $(x', y', z', w')$  three lineo-linear equations; writing these first under the form

$$\begin{aligned}(P_1, Q_1, R_1, S_1 \mid x', y', z', w') &= 0, \\ (P_2, Q_2, R_2, S_2 \mid x', y', z', w') &= 0, \\ (P_3, Q_3, R_3, S_3 \mid x', y', z', w') &= 0,\end{aligned}$$

we have  $x' : y' : z' : w' = X : Y : Z : W$ , where  $X, Y, Z, W$  are the determinants (each with its proper sign) formed out of the matrix

$$\begin{pmatrix} P_1 & Q_1 & R_1 & S_1 \\ P_2 & Q_2 & R_2 & S_2 \\ P_3 & Q_3 & R_3 & S_3 \end{pmatrix}.$$

106. Each of the surfaces  $X = 0, Y = 0, Z = 0, W = 0$ , or generally any surface  $aX + bY + cZ + dW = 0$ , is thus a cubic surface passing through the curve

$$\begin{vmatrix} P_1 & Q_1 & R_1 & S_1 \\ P_2 & Q_2 & R_2 & S_2 \\ P_3 & Q_3 & R_3 & S_3 \end{vmatrix} = 0,$$

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which is at once seen to be of the order 6. In fact any two of these surfaces, for instance

$$\begin{vmatrix} P_1 & Q_1 & R_1 \\ P_2 & Q_2 & R_2 \\ P_3 & Q_3 & R_3 \end{vmatrix} = 0 \quad \text{and} \quad \begin{vmatrix} P_1 & Q_1 & S_1 \\ P_2 & Q_2 & S_2 \\ P_3 & Q_3 & S_3 \end{vmatrix} = 0$$

have in common a curve

$$\begin{vmatrix} P_1 & Q_1 \\ P_2 & Q_2 \\ P_3 & Q_3 \end{vmatrix} = 0,$$

which is of the order 3; they consequently besides intersect in a curve of the order 6, which is the before mentioned curve of intersection of all the surfaces. And it further appears that the number of the apparent double points is  $= 7$ ; in fact the formula in the case of two surfaces of the orders  $\mu, \nu$ , the complete intersection of which consists of a curve of the order  $m$  with  $h$  apparent double points, and of a curve of the order  $m'$  with  $h'$  apparent double points, the numbers  $m, m', h, h'$  are connected by the equation  $m' - m - 2(h - h') = (m - \mu)(\nu - 1)$ . (Salmon's *Solid Geometry*, 2nd Ed., p. 273 [Ed. 4, p. 311]). Hence, in the case of the two cubic surfaces intersecting as above (since for the cubic curve we have  $m' = 3, h' = 1$ , and for the sextic  $m = 6$ ), the formula becomes  $2(h - 1) = 12$ , that is  $h = 1 + 6 = 7$ ; or the number of apparent double points is  $= 7$ .

107. It thus appears that the principal system in the first plane is a curve of the order 6, with seven apparent double points: it is to be added that there are not in general any actual double points or stationary points, so that the class of the curve is  $6 \cdot 5 - 2 \cdot 7 = 16$ , and its deficiency is  $\frac{1}{2} \cdot 5 \cdot 4 - 7 = 3$ . For convenience I will refer to this as the curve  $\Sigma$ .

The transformation is obviously a symmetrical one; hence the principal system in the second space is in like manner a curve of the order 6, with seven apparent double points; say it is the curve  $\Sigma'$ .

108. Consider in the first space any point  $P$  on the curve  $\Sigma$ ; for this point the three equations

$$\begin{aligned}(P_1, Q_1, R_1, S_1 \mid x', y', z', w') &= 0, \\ (P_2, Q_2, R_2, S_2 \mid &, \quad ) = 0, \\ (P_3, Q_3, R_3, S_3 \mid &, \quad ) = 0,\end{aligned}$$

are not independent, but are equivalent to two linear equations in  $(x', y', z', w')$ ; that is, to the point  $P$  on the curve  $\Sigma$  there corresponds in the second space, not a determinate point  $P'$ , but any point whatever on a certain line  $L'$ ; or say to the point  $P$  on  $\Sigma$  there corresponds a line  $L'$ ; and as  $P$  describes the curve  $\Sigma$ ,  $L'$  describes a scroll  $\Pi'$ ; that is, to the curve  $\Sigma$  there corresponds a scroll  $\Pi'$ , the principal counter-system in the second space. Similarly to the curve  $\Sigma'$  there corresponds a scroll  $\Pi$ , the principal counter-system of the first space.

109. The scroll  $\Pi$  is the Jacobian of the first space; and as such it is of the order 8, having the curve  $\Sigma$  for a triple line; and it thus appears that the Jacobian

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of the first space is a scroll (a theorem the analytical verification of which seems by no means easy). But without assuming the identity of the scroll  $\Pi$  with this Jacobian, or taking the order of the scroll to be known, I proceed to show that the scroll  $\Pi$  is the scroll generated by the lines each of which meets the curve  $\Sigma$  three times; it will thereby appear that the order is  $= 8$ , and that the curve is a triple line on the scroll.

Consider a point  $P'$  on  $\Sigma'$ , and the corresponding line  $L$  of the first space; take a plane  $\Theta'$  in the second space; corresponding to it the cubic surface  $\Theta$  in the first space. By imposing a single relation on the coefficients  $(a, b, c, d)$  in the equation  $ax' + by' + cz' + dw' = 0$  of the plane  $\Theta'$ , we make it pass through the point  $P'$ ; therefore by imposing this same single relation on the coefficients  $(a, b, c, d)$  of the cubic surface  $\Theta$ , we make it pass through the line  $L$ .  $\Theta$  is a cubic surface through  $\Sigma$ ; and it is easy to see that the effect will be as above only if the line  $L$  cuts the curve  $\Sigma$  three times; this being so, the general cubic surface  $\Theta$  meets  $L$  in three points (viz., the three intersections of  $L$  with  $\Sigma$ ), and if  $\Theta$  be made to pass through a fourth point on the line  $L$ , it will pass through the line  $L$ ; it thus appears that the line  $L$  meets  $\Sigma$  three times, and consequently that the scroll  $\Pi$  is generated by the lines which meet  $\Sigma$  three times.

110. The theory of a scroll so generated is considered in my “Memoir on Skew Surfaces, otherwise Scrolls”<sup>(1)</sup>. Writing  $m = 6$ ,  $h = 7$  and therefore  $M [= \frac{1}{2}m(m-1) - h]$ ,  $= 8$ , the order of the scroll is  $(\frac{1}{6}[m]^3 + (m-2)M = 40 - 32) = 8$ ; but calculating the values of

$$NG(m^2) = \frac{1}{8}[m]^4 + 6m + M(3[m]^2 - 12m + 33) + M^2 \cdot 3,$$

$$NR(m^2) = \frac{1}{30}[m]^5 + \frac{1}{4}[m]^4 - \frac{2}{3}[m]^3 - 3m + M(\frac{1}{2}[m]^3 - 4m^2 + 8m - 20) + M^2(\frac{1}{2}[m]^2 - 27m)$$

these are found to be respectively  $= 0$ ; viz., there are no nodal generators, and no nodal residue; the sextic curve  $\Sigma$  is a triple curve on the surface, and there is not any other multiple line.

111. It may be remarked that any plane  $\Theta'$  meets the sextic curve  $\Sigma'$  in six points; hence the corresponding cubic surface  $\Theta$  contains six lines, generatrices of  $\Pi$ , and, therefore, each meeting the curve  $\Sigma$  three times; say six lines  $L$ . Through one of these lines  $L$ , draw to the cubic surface  $\Theta$  a triple tangent plane meeting it in the line  $L$  and in two other lines, say  $M, N$ ; this plane must meet  $\Sigma$  in three new points which must lie on the lines  $M, N$ ; viz., one of these lines must pass through two of the points, and the other line through the third point.

*Addition — September, 1870.*

[Some corrections have been made in accordance with the concluding paragraph of a paper “Note on the Rational Transformation and on Special Systems of Points,” 450.]

The formulae of No. 84 are included in the following more general formulae; viz., if the principal system consist of  $\alpha_1$  points, each a simple point,  $\alpha_2$  points each a

<sup>1</sup> *Phil. Trans.* vol. CLIII. 1863, pp. 453–483, [339]. See the Table  $S(m^2)$  &c., p. 457; in the value of  $NR(m^2)$  instead of term  $+3m$  read  $-3m$ . [This correction should have been made in the present Reprint.]

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quadri-conical point,  $\alpha_3$  points each a cubi-conical point, &c., and of a simple curve order  $m_1$  with  $h_1$  apparent double points, a double curve order  $m_2$  with  $h_2$  apparent double points, and so on; and if moreover, the curves  $m_1, m_2$  intersect in  $k_{1,2}$  points, the curves  $m_1, m_3$  in  $k_{1,3}$  points, &c.; then writing in



general  $p = \frac{1}{2}m(m-1) - h$ ; that is,  $p_1 = \frac{1}{2}m_1(m_1-1) - h_1$ ,  $p_2 = \frac{1}{2}m_2(m_2-1) - h_2$ , &c., I find that the general condition of equivalence is

$$\left. \begin{aligned} &\alpha_1 + (3n-2)\alpha_2 \\ &\quad + 8\alpha_3 + (12n-16)\alpha_3 \\ &\quad + r^2\alpha_r + (3r^2n-2r^2)\alpha_r - 2r^2p_r \\ &\quad - 5k_{1,2} - 8k_{1,3} \dots - (3r-1)k_{1,r} \\ &\quad - 28k_{2,3} - \dots \\ &\quad - r^2s^2(r,s)k_{r,s} \ (s < r) \end{aligned} \right\} = n^2 - 1;$$

and that the general condition of postulation is

$$\left. \begin{aligned} &\alpha_1 + (n+1)\alpha_1 + p_1 \\ &\quad + \frac{1}{6}r(r+1)(r+2)\alpha_r \\ &\quad + [\frac{1}{2}r(r+1)n - \frac{1}{6}r(r+1)(2r-5)]m_r \\ &\quad - \frac{1}{24}[(r-1)(r-2)(r-3)(r-4) \\ &\quad \quad + 4r(r+1)(2r+1)]p_r \\ &\quad - \frac{5}{2}k_{1,2} - 5k_{1,3} \dots - rk_{1,r} \\ &\quad - 8k_{2,3} \\ &\quad - \frac{1}{6}r(r+1)(r+1 - \frac{1}{2}(r+2))k_{r,s} \ (s < r) \end{aligned} \right\} = \frac{1}{24}(n+1)(n+2)(n+3) - 4;$$

in which formulae it is however assumed that the curves have not any actual multiple points. This implies that if any one of the curves, say  $m_r$ , break up into two or more curves, the component curves do not intersect each other; for, of course, any such point of intersection would be an actual double point on the curve  $m_r$ . I believe, however, that the formulae will extend to this case by admitting for  $s$  the value  $s = r$ ; viz., if we suppose the curve  $m_r$  to be the aggregate of the two curves  $m'_r$ ,  $m''_r$  intersecting in  $\kappa_r$  points, then that the corresponding terms in the equivalence-equation are

$$\begin{aligned} &r^2(n^2 - (3rn - 2r^2)) \\ &\quad + r^2(m'_r + m''_r) - 2r^2(p'_r + p''_r) - r^2\kappa_r, \end{aligned}$$

and that those in the postulation-equation are

$$\begin{aligned} &- [\frac{1}{2}r(r+1)n - \frac{1}{6}r(r+1)(2r-5)](m'_r + m''_r) \\ &\quad + \frac{1}{24}[(r-1)(r-2)(r-3)(r-4) - 4r(r+1)(2r+1)](p'_r + p''_r) \\ &\quad - \frac{1}{2}r(r+1)(2r+1)\kappa_r. \end{aligned}$$

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Let the  $r$ -tuple curve consist of three right lines meeting in a point: this is an actual triple point, and the formulae do not apply. But calculating the postulation-terms by the formula, we have  $m_r = 3$ ,  $p_r = \frac{1}{2} \cdot 3 \cdot 2 - 0 = 3$ ; and the terms are

$$- [\frac{1}{2}r(r+1)n - \frac{1}{6}r(r+1)(2r-5)] \cdot 3 + \frac{1}{24}[(r-1)(r-2)(r-3)(r-4) + 4r(r+1)(2r+1)],$$

which are  $= \frac{1}{2}r(r+1)(3n-4r+4) - \frac{1}{24}(r-1)(r-2)(r-3)(r-4)$ , or say  $= \frac{1}{2}r(r+1)(3n-4r+4) + \frac{1}{24}(-r^4 + 10r^3 - 35r^2 + 50r - 24)$ .

I have found by an independent investigation that this value requires the correction

$$+ \frac{1}{24}[r^4 - 8r^3 + 30r^2 - 56r + 24 + \frac{1}{2}\{1 - (-)^r \cdot 1\}],$$

and that the true value of the postulation is

$$H = \frac{1}{2}r(r+1)(3n-4r+6) + \frac{1}{12}[2r^3 - 5r^2 - 6r + \frac{1}{2}\{1 - (-)^r \cdot 1\}],$$

viz., that this is the number of the conditions to be satisfied that a surface of the order  $n$  may have for an  $r$ -tuple curve three given right lines meeting in a point.

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NOTE ON THE CARTESIAN WITH TWO IMAGINARY AXIAL FOCI.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 181, 182. Read June 9, 1870.]

Let  $A, A', B, B'$  be a pair of points and antipoints; viz.,

$(A, A')$  the two imaginary points, coordinates  $(\pm\beta i, 0)$ ,

$(B, B')$  the two real points, coordinates  $(0, \pm\beta)$ ;

and write  $\rho, \rho', \sigma, \sigma'$  for the distances of a point  $(x, y)$  from the four points respectively; viz.,

$$\begin{aligned}\rho &= \sqrt{(x + \beta i)^2 + y^2}, & \sigma &= \sqrt{x^2 + (y + \beta)^2}, \\ \rho' &= \sqrt{(x - \beta i)^2 + y^2}, & \sigma' &= \sqrt{x^2 + (y - \beta)^2}.\end{aligned}$$

We have

$$\begin{aligned}\rho^2 + \rho'^2 &= 2x^2 + 2y^2 - 2\beta^2 = 2y^2 - \sigma^2 - \sigma'^2 + 4\beta^2, \\ \rho\rho' &= \sqrt{(x + \beta i + yi)(x + \beta i - yi)(x - \beta i + yi)(x - \beta i - yi)} = \sigma\sigma';\end{aligned}$$

and thence

$$\begin{aligned}(\rho + \rho')^2 &= (\sigma + \sigma')^2 - 4\beta^2, \\ (\rho - \rho')^2 &= (\sigma - \sigma')^2 - 4\beta^2,\end{aligned}$$

or say

$$\begin{aligned}\rho + \rho' &= \sqrt{(\sigma + \sigma')^2 - 4\beta^2}, \\ i(\rho - \rho') &= \sqrt{4\beta^2 - (\sigma - \sigma')^2}.\end{aligned}$$

The equation of a Cartesian having the two imaginary axial foci  $A, A'$  is

$$(p + qi)\rho + (p - qi)\rho' + 2k^2 = 0;$$

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viz., this is

$$p(\rho + \rho') + qi(\rho - \rho') + 2k^2 = 0;$$

or, what is the same thing, it is

$$p\sqrt{(\sigma + \sigma')^2 - 4\beta^2} + q\sqrt{4\beta^2 - (\sigma - \sigma')^2} + 2k^2 = 0,$$

which is the equation expressed in terms of the distances  $\sigma, \sigma'$  from the non-axial real foci  $B, B'$ . Of course, the radicals are to be taken with the signs  $\pm$ . This equation gives, however, the Cartesian in combination with an equal curve situate symmetrically therewith in regard to the axis of  $y$ .

The distances  $\sigma, \sigma'$  may conveniently be expressed in terms of a single variable parameter  $\theta$ ; in fact, we may write

$$\begin{aligned}\pm p\sqrt{(\sigma + \sigma')^2 - 4\beta^2} &= -k^2 - k\theta, \\ \pm q\sqrt{4\beta^2 - (\sigma - \sigma')^2} &= -k^2 + k\theta;\end{aligned}$$

that is

$$\begin{aligned}(\sigma + \sigma')^2 - 4\beta^2 &= \frac{k^2}{p^2}(k + \theta)^2, \\ 4\beta^2 - (\sigma - \sigma')^2 &= \frac{k^2}{q^2}(k - \theta)^2;\end{aligned}$$

and therefore

$$\sigma + \sigma' = \pm \sqrt{4\beta^2 + \frac{k^2}{p^2}(k + \theta)^2},$$

$$\sigma - \sigma' = \pm \sqrt{4\beta^2 - \frac{k^2}{q^2}(k - \theta)^2};$$

so that, assigning to  $\theta$  any given value, we have  $\sigma$ ,  $\sigma'$ , and thence the position of the point on the curve. We may draw the hyperbola  $y^2 = 4\beta^2 + \frac{k^2}{p^2}x^2$ , and the ellipse  $y^2 = 4\beta^2 - \frac{k^2}{q^2}x^2$ ; and then measuring off in these two curves respectively the ordinates which belong to the abscissae  $k + \theta$  for the hyperbola,  $k - \theta$  for the ellipse, we have the values  $\sigma + \sigma'$  and  $\sigma - \sigma'$ , which determine the point on the curve. Considering  $k$ ,  $p$ ,  $q$ ,  $\beta$  as disposable quantities, the conics may be any ellipse and hyperbola whatever, having a pair of vertices in common; and the complete construction is,—From the

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fixed point  $K$  in the axis of  $x$ , measure off in opposite directions the equal distances  $KM$ ,  $KN$ , and take

$$\frac{1}{2}(\sigma + \sigma') \text{ the ordinate at } M \text{ in the hyperbola,}$$

$$\frac{1}{2}(\sigma - \sigma') \text{ } N \text{ in the ellipse,}$$

where  $\sigma$ ,  $\sigma'$  denote the distances of the required point from the fixed points  $B$  and  $B'$  respectively, the distance of each of these from the origin being  $= \frac{1}{2}$  the common semi-axis. We may imagine  $N$  travelling from one extremity of the  $x$ -axis of the ellipse to the other, the value of  $\sigma + \sigma'$  will be real and greater than  $BB'$ , that of  $\sigma - \sigma'$  real and less than  $BB'$ , and the point  $(\sigma, \sigma')$  will be real. The construction gives, it will be observed, the two symmetrically situated curves.

The  $x$ -semi-axis of the ellipse is  $\frac{q}{k} \cdot 2\beta$ , and the form of the curve depends chiefly on the value of the ratio  $k : \frac{q}{k} \cdot 2\beta$ ; or, what is the same thing,  $k^2 : 2\beta q$ . We see, for instance, that, in order that the curve may meet the axis of  $y$  in two real points between the foci, the value  $\sigma - \sigma' = a - a'$ ; viz., that we must give a real value of must have  $\frac{k^4}{4\beta^2} > 4k^2$ ; that is,  $\beta^2 q^2 > k^2$ , or  $k^2 < 2\beta q$ . If  $k$  has this value, viz.,  $k = \frac{q}{k} \cdot 2\beta = \frac{1}{2}$  semi-axis, the curve touches the axis of  $y$  at the origin; if  $k < \frac{1}{2}$  semi-axis, the curve cuts the axis of  $y$  in two real points between the foci; if  $k > \frac{1}{2}$  semi-axis, the curve does not cut the axis of  $y$  between the foci.

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**SKETCH OF RECENT RESEARCHES UPON QUARTIC AND QUINTIC SURFACES.**

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 186–195. Read Nov. 10, 1870.]

The classification of quartic surfaces is even as to its highest divisions incomplete; and it is by no means easy to make it at once exhaustive and precise; an enumeration of all the *prima facie* possible cases would include forms which do not really exist. Thus the singular curve (if any) is of the order 1, 2, or 3; but in the case where the order is  $= 3$ , the curve, as is at once evident, cannot be a plane cubic, nor (among other excluded forms) a system of three non-intersecting lines. And certain forms of the singular curve, e.g. all but one of the admissible forms of a curve of the order 3, make the surface to be a scroll, so that, if (as is convenient) we wish to separate the scrolls, certain forms otherwise admissible must be excluded. The expression “singular” means double or cuspidal, or refers to a higher singularity, but the cases of higher singularity are very special. I will, at the cost of some inaccuracy, use the expression “nodal” as meaning, in general, double, but as including the signification “cuspidal”; and, if there are any cases of higher singularity, as extending to cases of higher singularity: and I provisionally arrange the non-scalar quartic surfaces as follows:

1. Without a nodal curve.
2. With a nodal line.
3. With a nodal conic, or line-pair (pair of intersecting lines).
4. With three nodal lines (not in the same plane) meeting in a point.

(Observe that the omitted cases are cases which, as I believe, ought to be omitted; thus the case of a nodal skew cubic is omitted, because the surface is then of necessity a scroll.) And to these I join

5. The quartic scrolls;

omitting altogether the torse and cones.

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The references, by the name of the author, and number (if any) of his paper, are to the subjoined list of Memoirs.

As to the scrolls, we have Cayley (3) and (4), and Cremona; the division into 12 species is, I believe, complete: see post, the remarks upon Schwarz's paper on quintic scrolls.

As regards the non-scalar surfaces:

1. Without a singular curve. The surface may be without a cnicnode (conical point), or it may have any number of cnicnodes up to 16, Cayley (7): the cases of singularity higher than a cnicnode are probably very numerous, but they have been scarcely at all examined. The memoir just referred to relates chiefly to the several cases of not more than 10 nodes; the cases of 11, 12, 13, 14, 15, 16 nodes are considered incidentally, Kummer (2), but it was not the object of his paper to make an enumeration, and there may be cases which are not considered; the discussion of the cases considered is very full and interesting. The case of 16 nodes is also considered, Kummer (1). As to the surface with 16 nodes, it is to be remarked that the wave-surface, or generally the surface obtained by the homographic deformation of the wave-surface called, Cayley (1), the "tetrahedroid"—is a special form of surface with 16 nodes: its relation to the general surface is explained, Cayley (2).

2. Quartic surface with nodal line: considered incidentally, Clebsch (2) and (3). There are through the nodal line 8 planes, each meeting the surface in a line-pair; considering any 7 of these, and taking out of each of them a line, the 7 lines are met by a conic which also meets a determinate line out of the remaining line-pair; there are thus on the surface  $2^7 = 128$ , conics; viz., these form 64 pairs, each pair lying in a plane, and being the complete intersection of the surface by such plane; the number of these planes is of course = 64.

Although not properly included in the present case, I mention the quartic surface which is the reciprocal of the cubic surface  $XIX = 12 - B_3 - C_2$ , Cayley (5): the nodal curve is here an oscnodal line counting as three nodal lines.

3. Quartic surfaces with nodal conic. Such a surface may be without cnicnodes, or it may have 1, 2, 3, or 4 cnicnodes; the cases, other than that of 3 cnicnodes, are mentioned, Kummer (3); but the question is examined, and the remaining case of 3 cnicnodes established, Cayley (6).

The general case of the nodal conic without cnicnodes is elaborately considered, Clebsch (1): it is shown that there are on the surface 16 lines, each meeting the conic, and which in their arrangement are strikingly analogous to the 27 lines on a cubic surface; viz., if on a cubic surface we select at pleasure any one of the 27 lines, and through this line draw a plane which besides meets the cubic surface in a conic; then, disregarding the line in question and the 10 lines which meet it, the remaining 16 lines each meet the conic, and are related to it and to each other in the same manner that the 16 lines of the quartic surface are related to the nodal

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conic and to each other. And the ground hereof appears, Geiser; viz., it is shown that the quartic surface with the nodal conic, is rationally transformable into a cubic surface, the 16 lines and the nodal conic corresponding respectively to the 16 lines and the conic of the cubic surface.

The several cases of 1, 2, 3, and 4 cnicnodes are considered, Korndorfer.

In the case where the nodal conic is the circle at infinity, the surfaces have been termed “anallagmatic” (perhaps “bicircular” would be a more convenient name), and a great deal has been written upon these surfaces by Moutard, Clifford, and others. Such a surface may of course have 1, 2, 3, or 4 cnicnodes; these surfaces, viz. the cnicnodal anallagmatics, in fact arise from the inversion of a quadric surface by the method of reciprocal radius vectors,—that is, by the change of  $x, y, z$  into  $\left(\frac{x}{r^2}, \frac{y}{r^2}, \frac{z}{r^2}\right)$ ; the centre of inversion is a node on the quartic surface. If the quadric surface is a cone, there is another node, the inverse point of the vertex; if the quadric surface is one of revolution, there are two other nodes; and if it is a cone of revolution, there are three other nodes—viz., in all, four nodes. The last-mentioned surface is, or includes, the Cyclide; viz., this is a quartic surface having the circle at infinity for a nodal curve, and having besides four nodes, which are a system of skew antipoints. The surface was first considered by Dupin (*Applications de Géométrie &c.*, 1822) as the envelope of a sphere touching three given spheres—its lines of curvature are thus circles; and the surface has been very frequently considered in reference to this property and otherwise: see Maxwell, where a classification (not quite complete) is made of the different forms of the surface, and also stereographic drawings given. It is to be observed, that one interesting form, the parabolic cyclide, is not a quartic but a cubic surface.

In the class of surfaces which have been under consideration, the cnicnodes have been points not on the nodal conic—in fact, a point on a nodal curve cannot be, properly speaking, a cnicnode, though it may be a point of higher singularity in the nature of a cnicnode; viz., there may be on the nodal curve points which, in the classification of the surfaces, must be counted as cnicnodes. Such a case presents itself in the “Conic Torus,” or surface generated by the rotation of a conic about a line whether not in or in the plane of the conic. The surface has been considered, De la Gournerie (1), although more in reference to the constructions of descriptive geometry than as a theory of pure geometry, and Cayley (6). The surface has a nodal circle, and upon it two singular points, the circular points at infinity; so that it belongs to the case of a nodal conic with two cnicnodes. In the particular case where the axis of rotation is in the plane of the conic, then there are on the axis two cnicnodes; so that the case is that of a nodal conic and four cnicnodes; and when the generating conic is a circle, viz., when the surface is the ordinary torus, or anchor ring, generated by the rotation of a circle about a line in its own plane, then the nodal conic is the circle at infinity having upon it two cnicnodal points (its intersections by the planes at right angles to the axis) and the surface has also two cnicnodes on the axis: the surface, although presenting considerable peculiarity, may be

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regarded as a particular case of the cyclide. In reference to the plane sections of the conic torus and its various particular cases, see De la Gournerie (2); the ordinary torus has been the subject of numerous papers by Darboux and others, and possesses very interesting properties.

In connexion with the foregoing, I speak of the surfaces having a cuspidal conic: the general case is briefly referred to, Cayley (6); viz., this is the surface (AA) the equation of which is  $V^2 - x^2y = 0$ , and which it is shown has a reciprocal of the order 6. A special case is the surface (AB) having a reciprocal of the order 3; viz., the quartic surface is here the reciprocal of the cubic surface  $XX = 12 - U_8$ , Cayley (5). And it appears from the memoir last referred to, that there is another cubic surface,  $XVII = 12 - 2B_3 - C_2$ , the reciprocal of which is a quartic surface having a cuspidal conic. But the theory of the quartic surfaces with a cuspidal conic has been hardly at all considered.

I do not know that anything has been done in regard to the quartic surfaces where the nodal conic becomes a line-pair; that is, where we have two intersecting nodal lines. Although not properly belonging to the case in question, I mention here the quartic surface which is the reciprocal of the cubic surface  $XVIII = 12 - B_4 - 2C_2$ , Cayley (5); the nodal curve consists of two intersecting lines, but one of them is tacnodal, counting as two nodal lines.

4. Quartic surface with three nodal lines (not in the same plane) meeting in a point. This is, in fact, Steiner's quartic surface; and it has been the subject of numerous investigations.

The equation of the surface may be taken to be  $\sqrt{x} + \sqrt{y} + \sqrt{z} + \sqrt{w} = 0$ ; and the surface thus presents itself as the reciprocal of the cubic surface  $\frac{1}{w} + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$ , ( $XVI = 12 - 4C_2$ ) with four cnicnodes<sup>(1)</sup>.

It was convenient to make the foregoing enumeration before speaking of Kummer's paper (3), and of the several memoirs which relate to the Abbildung of certain quartic and quintic surfaces.

As regards Kummer's paper, the object appears by the title, viz., he considers in what cases a quartic

surface has upon it a system of conics; or, what is the same thing, in what cases there is a system of planes each intersecting the surface in two conics. It is, in the first place, remarked that there is no proper quartic surface cut by every plane in a pair of conics, or even a proper quartic surface cut in a pair of conics by every plane through a fixed point. The cases considered are I., where the planes are non-tangent planes; II., where they are single tangent planes; and III., where they are double tangent planes. The case I. is—(1) when there is a nodal conic and two cnicnodes; viz., any plane through the 2 cnicnodes gives a section with 4 nodes, therefore a pair of conics, (and the special case of 4 cnicnodes is noticed

<sup>1</sup> The Author exhibited, and pointed out some of the properties of, a model of Steiner's surface.

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incidentally); (2) when there is a nodal line; any plane through the nodal line besides meets the surface in a conic; (3) when the surface has two “Selbstberührungspuncte”—viz., either of these is a point where the tangent plane is replaced by two coincident planes, and which, when a plane passes through it, gives in the section a tacnode, = 2 nodes; the section by a plane through two such points, consists of two conics touching each other at the point in question: the equation of such a surface is  $\phi^2 = (* | p, q)^2$ , where  $\phi$  is a quadric function, and  $p, q$  linear functions of the coordinates. In all the cases the planes pass through a fixed line, and the surface may be considered as the locus of a variable conic, the plane of which always passes through such line. II. is (1) Steiner's surface, where every tangent plane meets the surface in a pair of conics; and (2) surface with a nodal conic and one cnicnode, where every tangent plane through the cnicnode meets the surface in a pair of conics. And III. is (1) the surface with a nodal conic, where every double tangent plane meets the surface in a pair of conics: it is shown that there are 5 quadric cones, such that a tangent plane of any one of these cones is always a double tangent plane of the surface. Or the surface is (2) a quartic scroll; any plane through two intersecting lines of the surface besides meets the surface in a conic.

It is in the paper, Cayley (6), remarked, that the quartic surface  $(* | U, V, W)^2 = 0$  can also be expressed in the form  $UW - V^2 = 0$ ; under which form the surface is seen to be the envelope of the series of quadric surfaces  $(U, V, W | \theta, 1)^2 = 0$ ; and by reason of this property it is very easy to find the equations of the reciprocal surfaces, or plane-equations of the quartic surfaces in question. And, in the same paper, it is noticed that the surfaces of the form in question include the reciprocals of several interesting surfaces of the orders 6, 8, 9, 10, and 12; viz., order 6, parabolic ring; order 8, elliptic ring; order 9, centro-surface of paraboloid; order 10, parallel surface of paraboloid; envelope of planes through the points of an ellipsoid at right angles to the radius vectors from the centre; order 12, centro-surface of ellipsoid; parallel surface of ellipsoid.

It will be noticed that several of the papers by Clebsch and others refer in their titles to the “Abbildung” of a surface; viz., they show that a (1, 1) correspondence exists between the points of the surface and the points of a plane. The most simple instance is the quadric surface; here, taking any fixed point  $O$  on the surface, the line  $OP$  drawn to any point  $P$  on the surface meets a plane in a point  $P'$ , and the points  $P, P'$  have, it is clear, a (1, 1) correspondence. And, of course, to any curve on the quadric surface there corresponds a curve on the plane, and the discussion of the nature of the plane curves which correspond to the different curves on the quadric surface would constitute a theory of the Abbildung of the quadric surface.

Similarly, as remarked, Clebsch (2), for a cubic surface, taking upon it any two lines which do not meet, if from a point  $P$  on the surface we draw, meeting each of the two lines, a line to meet the plane in  $P'$ , then the point  $P$  on the cubic surface and the point  $P'$  on the plane will have a (1, 1) correspondence; and we have thus a like theory for the cubic surface. The Abbildung of a cubic surface had been, however, previously effected by Clebsch {in the paper “Die Geometrie auf den Flächen

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dritter Ordnung,” *Crelle*, t. LXV. (1866), pp. 359–380}, and by Cremona, in a different and really the most simple manner (<sup>1</sup>), but having a less obvious geometrical signification.

For surfaces of the higher orders, it is only certain surfaces which admit of an Abbildung, or (1, 1) correspondence of the points thereof with the points of a plane; viz. (in the same way as a plane curve, in order to its being unicursal, must have a sufficient number of nodes or cusps) a surface, in order that it may thus correspond with the plane (or say, in order that it may be unicursal), must have a sufficient

singularity in the way of a nodal or cuspidal curve. The quartic and quintic surfaces considered in the enumerated memoirs are there considered for the sake of the Abbildung theory which they give rise to; whereas, in the present sketch, the Abbildung theory is considered only for the sake of the quartic and quintic surfaces to which the theory has been applied. But the methods of the theory furnish results in relation to these surfaces; and it is proper to give some account of them.

Clebsch's memoirs (2) and (3) relate to the same subject, which is elaborately treated in the latter of them: the former of them contains, however, some valuable remarks which are not reproduced in the other. In these memoirs (2) and (3), after explaining the above method of the transformation of a cubic surface by means of two of the lines thereof, the author goes on to notice that the like method is applicable to certain quartic and quintic surfaces; viz., (1) quartic surface with a nodal conic: there are here, as already mentioned, 16 lines, each meeting the conic; if, selecting any one of these, from a point  $P$  on the surface we draw, meeting the line and the conic, a line to cut the plane in  $P'$ , then the points  $P$  on the surface and  $P'$  on the plane have a (1, 1) correspondence. (2) Quartic surface with a nodal line: as already mentioned, there are on the surface 128 conics, each meeting the nodal line; selecting any one of these, if from a point  $P$  of the surface we draw, meeting the nodal line and the conic, a line cutting the plane in  $P'$ , then the point  $P$  on the surface, and the point  $P'$  on the plane, have a (1, 1) correspondence.

Similarly, (3), for a quintic surface having a nodal skew cubic; then if from a point  $P$  on the surface we draw, meeting the skew cubic twice, a line to cut the plane in  $P'$ , the point  $P$  on the surface and the point  $P'$  on the plane have a (1, 1) correspondence. The nodal skew cubic may break up into a conic and line which meets it, or into three lines, two of them not meeting each other, but each met by the third line; and the like theory applies to these quintic surfaces.

It is to be noticed that (as for the cubic surface) the above methods of Abbildung, although they have the most obvious geometrical significations, are (as explained in the foregoing foot-note) not the most simple ones; but for each of the foregoing cases (1), (2), (3), the most simple transformation is established in the memoirs now under consideration. The memoir [Memoirs (1) and (2)] of Korndorfer, as indicated by its title, relates to the Abbildung of a quartic surface having a nodal conic and 1, 2, 3, or 4 cnicnodes.

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<sup>1</sup> Any method of transformation leads to an expression of the coordinates of a point on the surface as proportional to rational and integral functions of a given degree  $\nu$  of the coordinates  $(x, y, z)$  of a point on the plane, and that transformation is the more simple for which  $\nu$  has the smaller value: for the method of the text, the value is  $\nu = 3$ , but for the methods previously given by Clebsch and Cremona, it is  $\nu = 2$ .

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Clebsch's paper (4) relates to the Abbildung of a quartic scroll.

As regards quintic surfaces (not being scrolls), we have, so far as I am aware, only the before-mentioned paper, Clebsch (3), relating to quintic surfaces with a nodal skew cubic; and the paper, Clebsch (5), which relates to the Abbildung of a quintic surface having a nodal quadric. The method employed is that of a preliminary Abbildung upon a twofold plane (2-blättrige Ebene); that is, it consists in establishing, in the first instance, a (1, 2) correspondence between the surface and the plane; and by means hereof it is shown that there exist on the surface the conics  $K$  and  $G$  presently referred to, and which give, ultimately, an ordinary Abbildung or (1, 1) correspondence of the points of the surface with those of the plane; viz., this final result is as follows:

There is on the surface a system of conics  $K$ , such that their planes pass through a point and envelope a quadricone; and also 64 conics  $G$  each meeting each of the conics  $K$  in a single point: we select one of these and call it the conic  $G$ .

Take now the plane  $B_1$  of a conic  $K_1$  of the series of conics  $K$ , which plane  $B_1$  passes, of course, through the vertex  $V$  of the cone enveloped by the planes of the conics  $K$ ; viz., these planes intersect the plane  $B_1$  in a series of lines passing through the point  $V$ .

Take a point  $P$  on the surface; this lies on a conic  $K$  meeting the conic  $G$  in a point  $\xi$ ; and if we draw the line  $\xi P$  to meet the plane  $B_1$  in  $P'$ , then  $P$  on the quintic surface, and  $P'$  on the plane  $B_1$ , will have a (1, 1) correspondence; in fact, it appears that, given  $P$ , there exists a single position of  $P'$ ; and conversely, given  $P'$ , this lies on a line  $VP'$  through which there passes the plane  $B_1$  and one other tangent plane of the cone: this tangent plane contains a conic  $K$  meeting the conic  $G$  in a point  $\xi$ ; and joining  $\xi P'$ , this

meets the conic  $K$  in one other point; viz., given  $P'$ , there is a single position of  $P$ ; and there is thus a (1, 1) correspondence.

There are, as originally shown by Schläfli, and as further appears by my memoir on cubic surfaces, Cayley (5), 3 kinds of cubic surfaces of the class 5; viz., these are the surfaces  $XIII = 12 - B_3 - 2C_2$ ,  $XIV = 12 - B_4 - C_2$ , and  $XV = 12 - U_7$ ; for each of these the reciprocal surface is a quintic surface of the class 3, having a nodal line and a cuspidal quartic curve. For the reciprocal of  $XIII$ , the cuspidal curve is a quadriquadric; for that of  $XIV$ , the cuspidal curve breaks up into the nodal line (viz., this is a cuspnodal line) and into a skew cubic; for that of  $XV$ , the cuspidal curve is a cuspidal quadriquadric, or curve of intersection of two quadric surfaces with singular contact.

It only remains to speak of Schwarz's memoir on quintic scrolls: it is to be remarked that the theory of scrolls is allied more closely with that of plane curves than with that of surfaces; viz., considering any plane section of the scroll, the lines of the scroll have, in general, a (1, 1) correspondence with the points of the plane section, and the scrolls of any given order are properly arranged according to the deficiency of the plane section. This is what is done by Cremona in the memoir on quartic

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scrolls above referred to; viz., for a quartic scroll the deficiency is either 0 or 1; and of the 12 species, there are 10 for which the deficiency is = 0 (or which are unicursal), and 2 for which the deficiency is = 1. And this is the principle of classification in Schwarz's memoir; viz., for a quintic scroll the deficiency is = 0, 1, or 2; the number of species established being 10, 4, and 1 for these deficiencies respectively.

#### *List of Memoirs.*

- Cayley.** 1. Sur la surface des ondes. *Liouv.* t. XI. (1846), pp. 291–296, [47].
2. Sur un cas particulier de la surface du quatrième ordre avec seize points singuliers. *Crelle*, t. LXV. (1866), pp. 284–290, [356].
3. Second Memoir on Skew Surfaces, otherwise Scrolls. *Phil. Trans.*, vol. CLIV. (1864), pp. 559–577, [340].
4. Third Memoir on Skew Surfaces, otherwise Scrolls. *Phil. Trans.*, vol. CLIX. (1869), pp. 111–126, [410].
5. Memoir on Cubic Surfaces. *Phil. Trans.*, vol. CLIX. (1869), pp. 231–326, [412].
6. On the Quartic Surfaces  $(* | U, V, W)^2 = 0$ . *Quart. Math. Journ.* vol. X. (1868), pp. 24–34; and vol. XI. (1870), pp. 15–25, and pp. 111–113.
7. Memoir on Quartic Surfaces. *Proc. Lond. Math. Soc.*, vol. III. (1870), pp. 19–69, [445].
- Clebsch.** 1. Ueber die Flächen vierter Ordnung welche eine Doppelcurve zweiten Grades besitzen. *Crelle*, t. LXIX. (1868), pp. 355–358.
2. Intorno alla rappresentazione di superficie algebriche sopra un piano. *Atti del R. Ist. Lomb.* (12 Nov. 1868), 13 p.
3. Ueber die Abbildung algebraischer Flächen, insbesondere der vierten und fünften Ordnung. *Math. Ann.*, t. I. (1869), pp. 253–316.
4. Ueber die ebene Abbildung der geradlinigen Flächen vierter Ordnung, welche eine Doppelcurve dritten Grades besitzen. *Math. Ann.*, t. II. (1870), pp. 445–466.
5. Ueber die Abbildung einer Classe von Flächen fünfter Ordnung. *Gött. Abh.*, t. XV. 64 p.
- Cremona.** Sulle superficie gobbe di quarto grado. *Mem. di Bologna*, t. VIII. (30 April, 1868), 15 p.
- De la Gournerie.** 1. Mémoire sur la surface engendrée par la révolution d'une conique autour d'une droite située d'une manière quelconque dans l'espace. *Jour. de l'Éc. Polyt.*, t. XXIII. (1863), pp. 1–74.
2. Mémoire sur les lignes spiriques. *Liouv.*, t. XIV. (1869), 92 p.



**Geiser.** Ueber die Flächen vierten Grades welche eine Doppelcurve zweiten Grades haben. *Crelle*, t. LXX. (1869), pp. 249–257.

**Korndorfer.** [There was only a general reference; the several memoirs are:

1. Die Abbildung einer Fläche vierter Ordnung mit einer Doppelcurve zweiten Grades und einem oder mehreren Knotenpunkten. *Math. Ann.*, t. I. (1869), pp. 592–626.
2. Fortsetzung dieses Aufsatzes. t. II. (1870), pp. 41–64.
3. Ueber diejenigen Raumcurven deren Coordinaten sich als rationale Functionen eines Parameters darstellen. t. III. (1871), pp. 415–423.
4. Die Abbildung einer Fläche vierter Ordnung mit zwei sich schneidenden Doppelgeraden. t. III. (1871), pp. 496–522.
5. Die Abbildung einer Fläche vierter Ordnung mit einer Doppelcurve zweiten Grades welche aus zwei sich schneidenden unendlich nahen Geraden besteht. t. IV. (1871), pp. 117–134.]

**Kummer.** 1. {Surfaces of the fourth order with sixteen conical points.} *Berl. Monatsb.* (1864), pp. 246–260, and 495–499.

2. Ueber die algebraischen Strahlensysteme, insbesondere über die der ersten und zweiten Ordnung. *Berl. Abh.* (1866), pp. 1–120.

3. Ueber die Flächen vierten Grades auf welchen Schaaren von Kegelschnitten liegen. *Berl. Monatsb.* (July, 1868). *Crelle*, t. LXIV. (1864), pp. 66–76.

**Maxwell.** On the Cyclide. *Quart. Math. Journ.*, t. IX. (1867), pp. 111–126.

**Schwarz.** Ueber die geradlinigen Flächen fünften Grades. *Crelle*, t. LXVII. (1867), pp. 23–57.

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### NOTE ON THE THEORY OF THE RATIONAL TRANSFORMATION BETWEEN TWO PLANES, AND ON SPECIAL SYSTEMS OF POINTS.

[From the *Proceedings of the London Mathematical Society*, vol. III. (1869–1871), pp. 196–198. Read December 8, 1870.]

In Prof. Cremona's theory of the transformation of plane curves, the fundamental equations are taken to be

$$\alpha_1 + 4\alpha_2 + 9\alpha_3 + \dots = n^2 - 1, \quad (1)$$

$$\alpha_1 + 3\alpha_2 + 6\alpha_3 + \dots = \frac{1}{2}(n^2 + 3n) - 2; \quad (2)$$

and from these we have as a consequence

$$\alpha_1 + 3\alpha_2 + \dots = \frac{1}{2}(n-1)(n-1); \quad (3)$$

viz., the first equation expresses that any two curves of the system intersect in a single variable point; the second, that the curves form a réseau, or system containing two arbitrary parameters; and the third, that the curves are unicursal.

In the equivalent theory of the rational transformation between two planes, as given in my "Mémor on the Rational Transformation between Two Spaces," [447], we have the equation (1); but instead of the equation (2), it would *primâ facie* appear to be sufficient if we had the inequation

$$\alpha_1 + 3\alpha_2 + 6\alpha_3 + \dots \leq \frac{1}{2}(n^2 + 3n) - 2$$

but on the ground there explained, the case

$$\alpha_1 + 3\alpha_2 + 6\alpha_3 + \dots < \frac{1}{2}(n^2 + 3n) - 2$$

is excluded, and we thus have the equation (2), giving with (1) the equation (3).